



PIONEER JUNIOR COLLEGE
JC2 PRELIMS EXAMINATIONS 2013

**Candidate
Name**

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**Civics
Group**

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**INDEX
NUMBER**

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H2 BIOLOGY

9648 / 01

27 September 2013

1 hour 15 mins

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number on the cover page in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the OTAS answer sheet provided.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Do not open this booklet until you are told to do so.

This document consists of **23** printed pages (inclusive of this page)

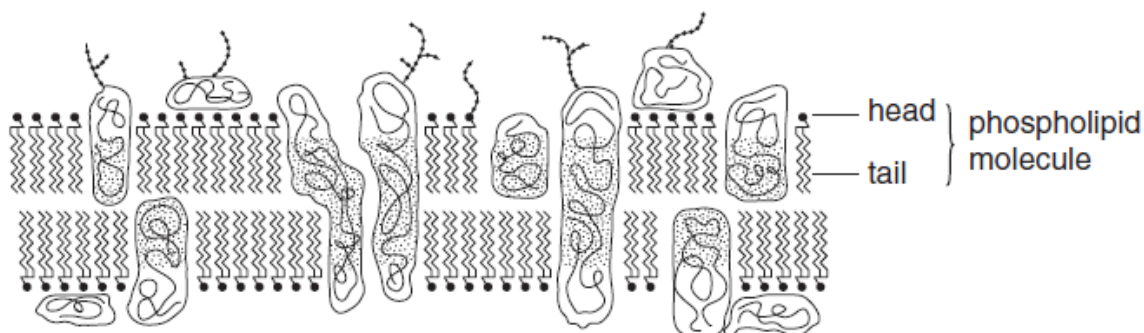
[Turn Over]

1. The table gives description of four membranous structures in a cell.

Which structure is correctly matched to its function?

	Structure	Function
A	an extensive network of tubes and sacs; each tube and sac bounded by a single membrane	Lipid synthesis
B	a spherical sac bounded by a single membrane	Protein synthesis
C	a sac bounded by two membranes, the inner highly folded	Packaging of proteins
D	a stack of elongated, curved sacs; each sac bounded by a single membrane	Photosynthesis

2. The diagram shows a section of a cell surface membrane.



What causes the phospholipid molecules to be arranged as shown?

- A** The heads are hydrophilic and point towards water molecules.
B The heads are hydrophobic and point away from water molecules.
C The tails are hydrophilic and point away from water molecules.
D The tails are hydrophobic and point towards water molecules.
3. Which carbohydrate has glycosidic bonds between carbon atom 1 and carbon atom 6 of its component glucose molecules?
- A** amylose
B glycogen
C maltose
D sucrose

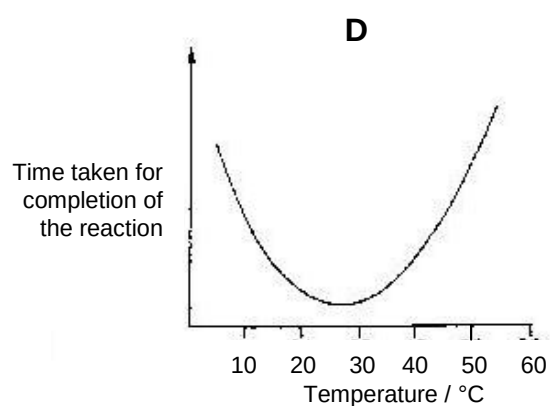
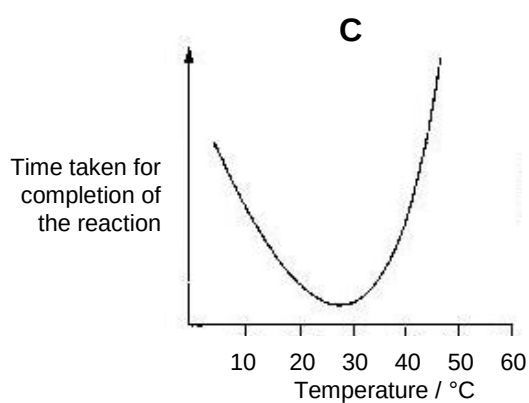
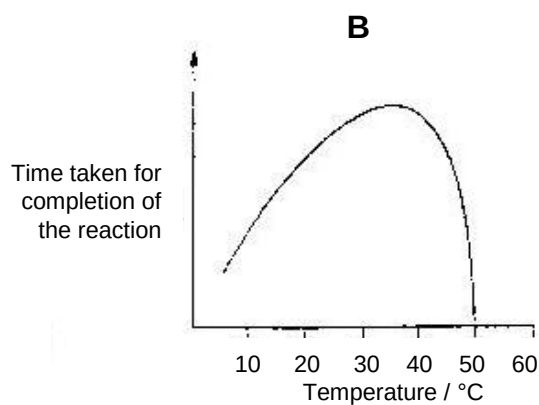
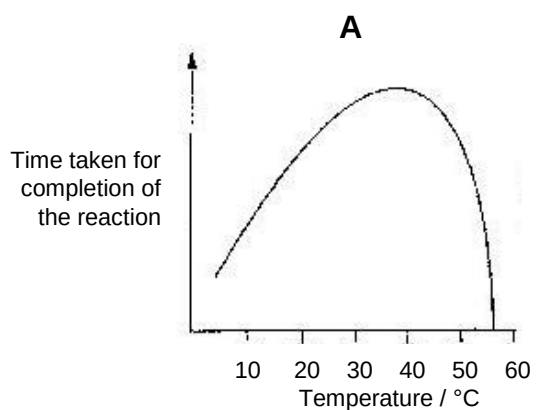
4. Compared to globular proteins, fibrous proteins are

- I. more regular in structure
- II. mostly involved in metabolic activities
- III. more readily soluble
- IV. more resistant to high temperature

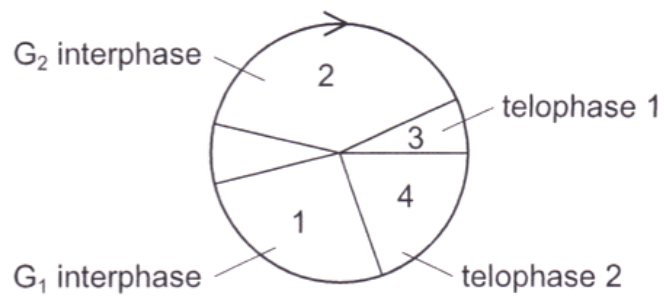
- A** I and II only
- B** I and IV only
- C** II and III only
- D** I, III and IV only

5. In an investigation to determine the effect of temperature on activity of an enzyme, the time for all the substrate to disappear from a standard solution was recorded.

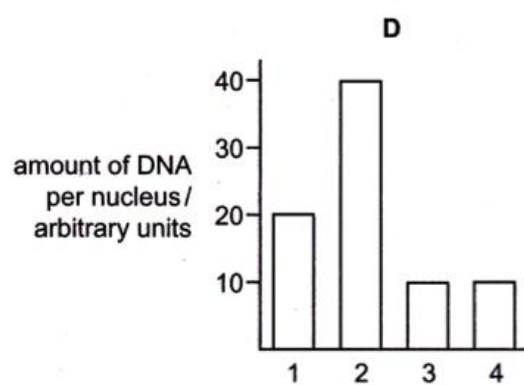
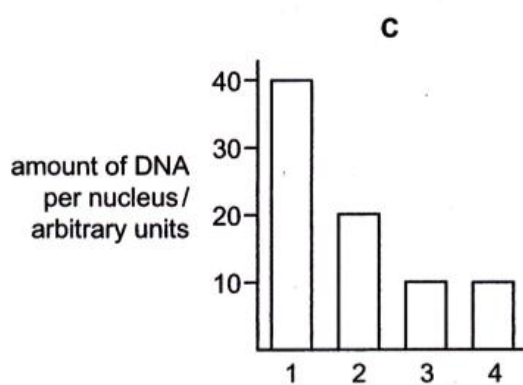
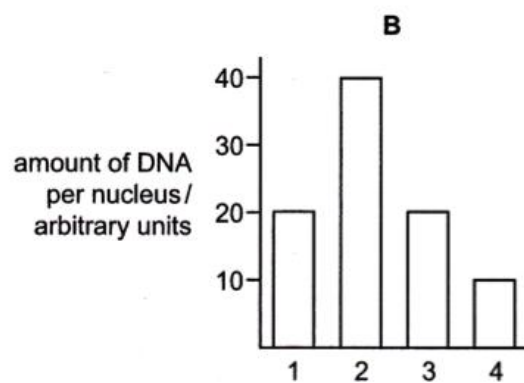
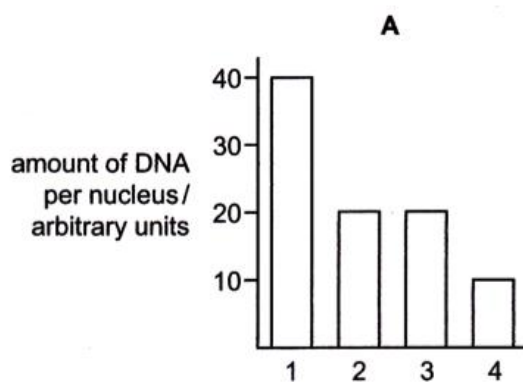
Which graph shows the results of this investigation?



6. The amount of DNA per nucleus in a cell during a meiotic cell cycle as shown below was measured.



Which bar correctly represents the variation in DNA content?



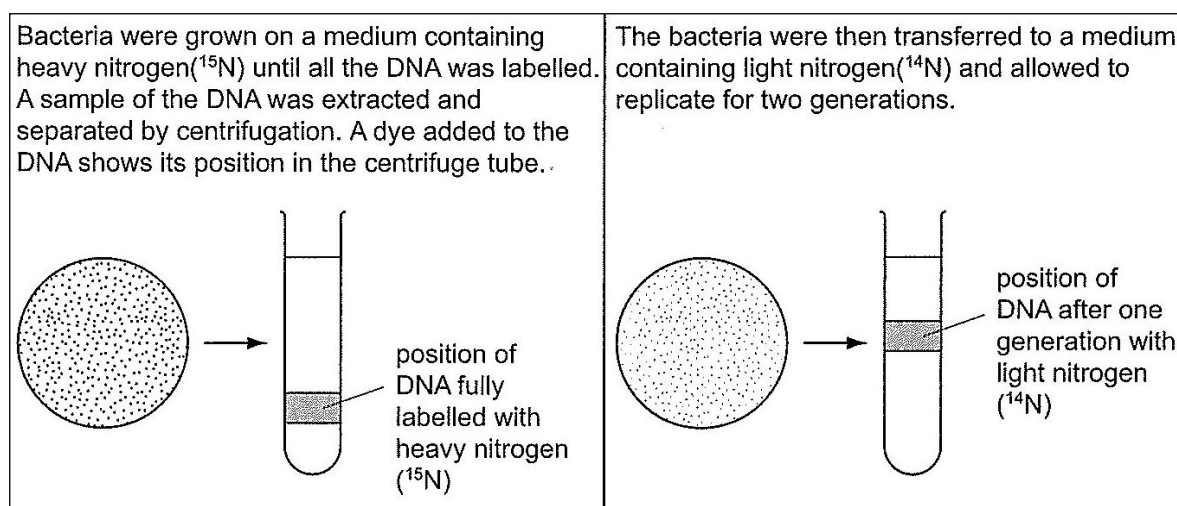
7. The packing of DNA in the nucleus is a very precise process. The following statements describe this process.

- I. DNA winds twice around a histone to form nucleosome.
- II. Coiling results in formation of condensed chromatin as seen in metaphase.
- III. Looped domains are formed with the aid of chromosome scaffold.
- IV. Coiling results in formation of a solenoid fibre.

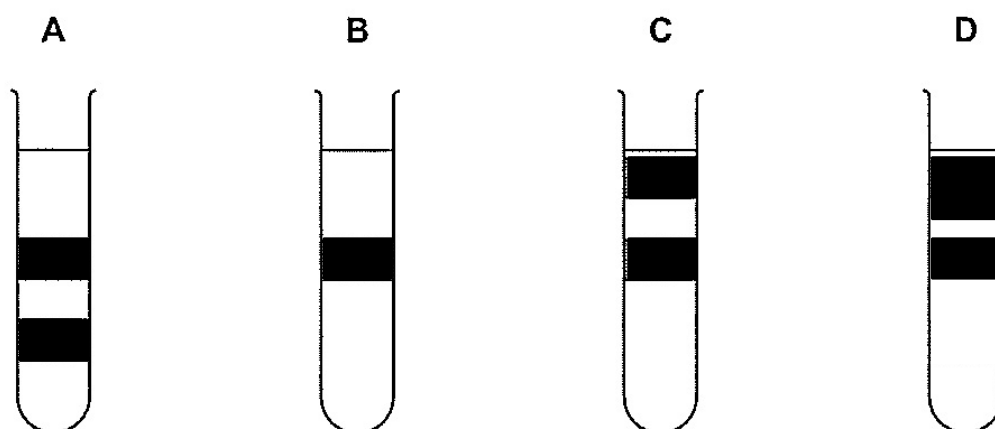
Which of the following correctly describes the sequence of DNA packing?

- A** I, III, IV, II
- B** I, IV, III, II
- C** III, I, IV, II
- D** III, IV, I II

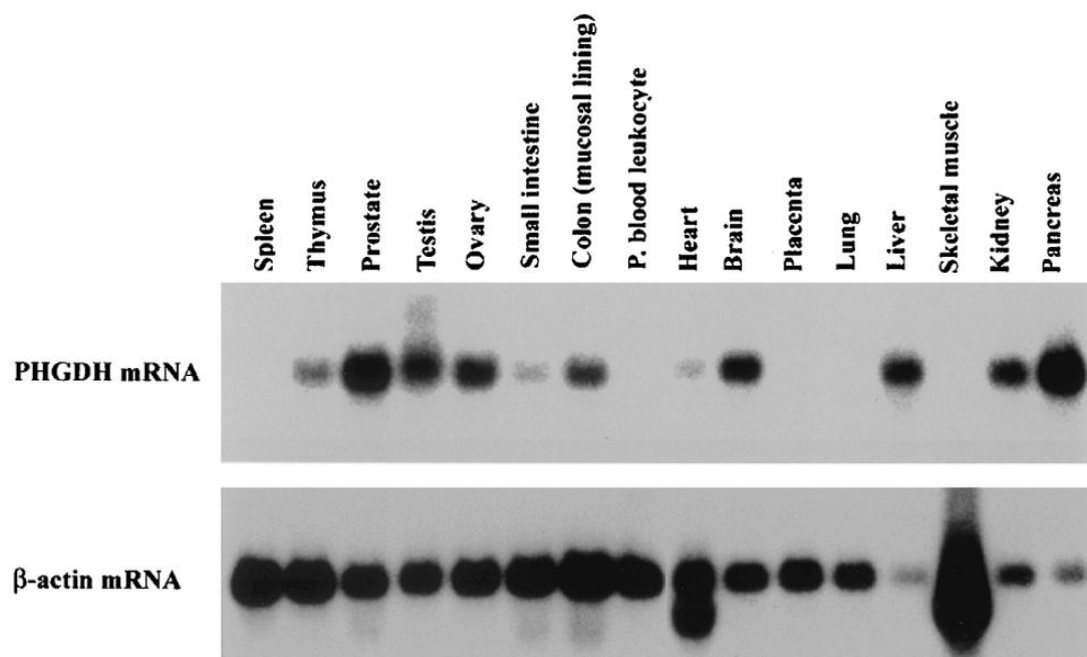
8. The diagrams show an investigation into semi-conservative replication of DNA.



Which tube shows the position of the DNA after two more generations of semi-conservative replication in light nitrogen (^{14}N)?



9. Which of the following is not involved in regulating the synthesis of RNA in the eukaryotic nucleus?
- A DNA methylation of DNA regions where genes are active.
 - B Amplification of some genes such as rRNA genes.
 - C Spliceosomes that splice introns from pre-mRNA to form mature mRNA.
 - D Use of different RNA polymerases to transcribe different classes of RNA.
10. Both β -actin and D-3-phosphoglycerate dehydrogenase (PHGDH) are proteins that can be found in the human body. Multiple tissues from the same individual were taken, followed by the isolation of the respective mRNA from the same number of cells of each tissue type. The mRNA were then subjected to gel electrophoresis. The following autoradiograph shows the result of this study.



From the figure above, which conclusion cannot be deduced from these results?

- A The gene that codes for PHGDH cannot be found in all types of cells.
- B Activators are bound to enhancers of the gene coding for β -actin in skeletal muscle cells.
- C The gene coding for β -actin is located in euchromatin.
- D Inhibitors are bound to silencers of the gene coding for PHGDH in lung cells.

11. Which statement defines control elements?

- A** A segment of DNA where transcription factors bind preferentially.
- B** A short region of DNA that can bind with proteins to enhance transcription levels.
- C** DNA sequences that interact with regulator proteins to determine the rate and timing of gene expression.
- D** Proteins found only in eukaryotes that regulate transcription by binding to DNA sequences to control the transfer of genetic information from DNA to mRNA.

12. Which of the following correctly describes the function of components involved in translation?

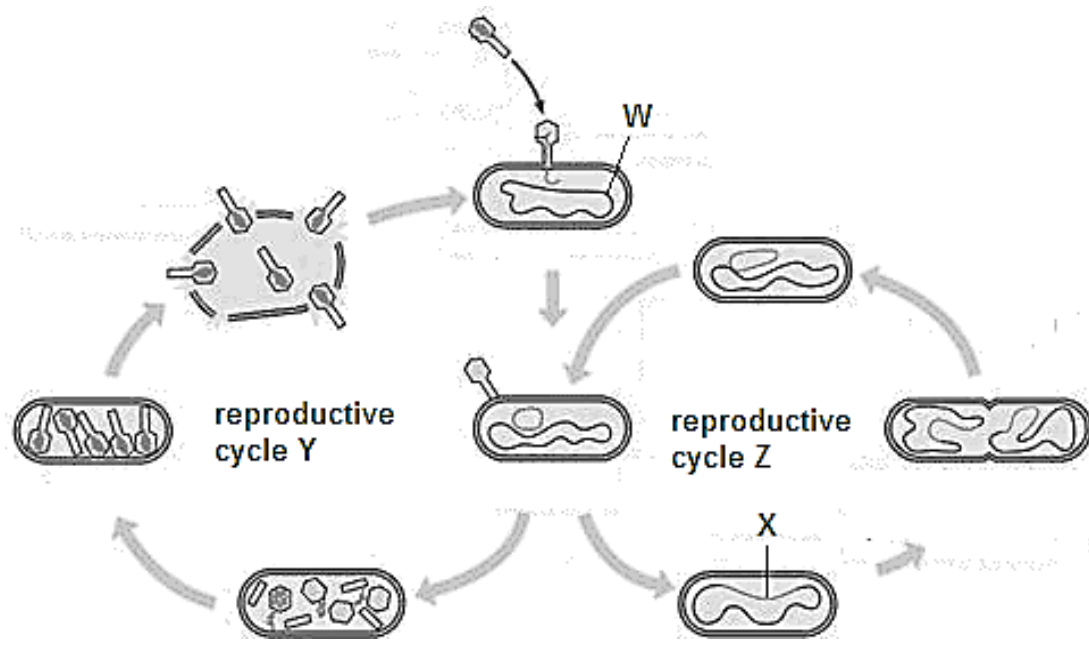
	Peptidyl transferase	Ribosomes	Aminoacyl-tRNA synthetase
A	Transfers an amino acid to a growing polypeptide chain	Contains an aminoacyl site that is occupied by the initiator tRNA attached to methionine	Attaches an amino acid to the 5' end of a tRNA
B	Forms a peptide bond between two amino acids	Contains a peptidyl site that is occupied by the initiator tRNA attached to methionine	Attaches an amino acid to the 5' end of a tRNA
C	Forms a peptide bond between two amino acids	Contains a peptidyl site that is occupied by the initiator tRNA attached to methionine	Attaches an amino acid to the 3' end of a tRNA
D	Transfers an amino acid to a growing polypeptide chain	Contains an aminoacyl site that is occupied by the initiator tRNA attached to methionine	Attaches an amino acid to the 3' end of a tRNA

13. A point mutation can occur anywhere on a chromosome. Which statements are true?

- I. Point mutation in an exon can alter the codon resulting in a different amino acid sequence.
- II. Point mutation in an exon can produce a shorter protein as a stop codon is produced.
- III. Point mutation in an intron can alter the binding site of a splicing enzyme.

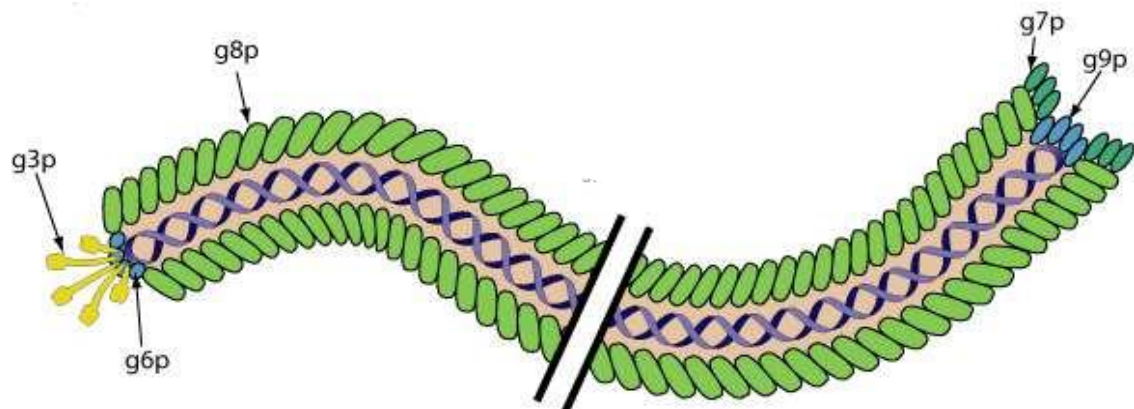
- A** I and II
- B** I and III
- C** II and III
- D** I, II and III

14. The following figure illustrates the reproductive cycle of a virus.



	W	X	Y	Z
A	viral DNA	prophage	lytic	lysogenic
B	viral RNA	provirus	lysogenic	lytic
C	bacterial DNA	provirus	lysogenic	lytic
D	bacterial DNA	prophage	lytic	lysogenic

15. The diagram below shows the structure of M13 bacteriophage. It has single-stranded circular DNA genome. g3p, g6p, g7p, g8p and g9p are capsid proteins. Its life cycle does not cause the lysis of the host cell.

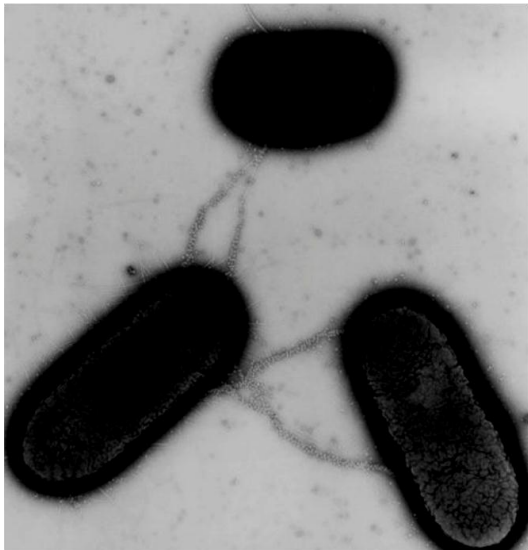


Based on your understanding of bacteriophages, which of the following statements are true of the M13 bacteriophage?

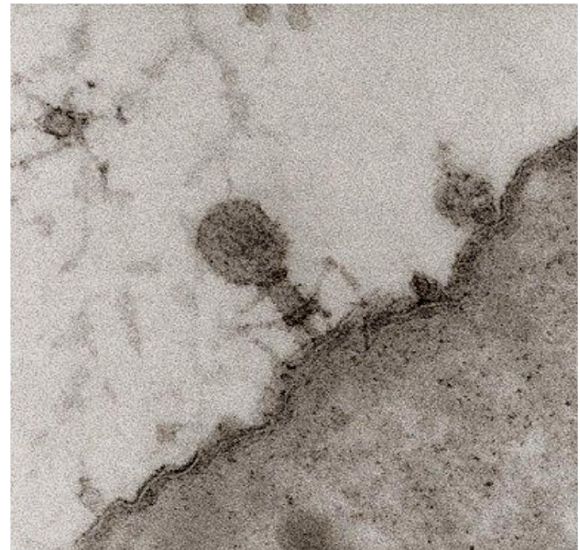
- I. The base composition of its genome is such that the relative ratio of C:G is 1:1.
 - II. At least one of the capsid proteins is responsible for binding to a specific protein on the host cell.
 - III. Its genome is injected into the host cell after the phage has attached to the host cell.
 - IV. It acquires its envelope from the cell membrane of its host cell.
- A** I and III
B II and III
C I, II and III
D II, III and IV

16. The photomicrographs below show two different processes of genetic transfer in bacteria.

Process R



Process S



A series of statements concerning the two processes is shown below.

- I. Physical contact between the donor and recipient bacterium is necessary.
- II. The donor bacterium is lysed in the process.
- III. Specific genes in the donor bacterium are required for the transfer to take place.
- IV. Any gene on the bacterial chromosome can be transferred.

Which row shows the correct statements for the corresponding process?

	Process R	Process S
A	I and II	III and IV
B	I and III	II and IV
C	II and III	I and IV
D	III and IV	I and II

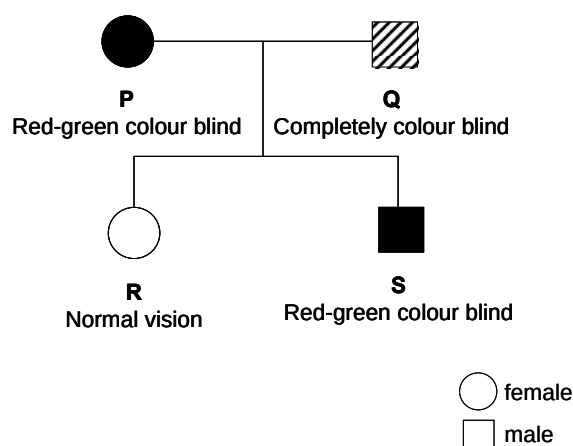
17. In prokaryotes, arginine operon is a negative repressible operon. A repressor, coded by a regulatory gene, controls the operon. Which of the following is correct?

	Absence of arginine	Presence of arginine
A	Repressor has high affinity for operator	Arginine binds to the repressor, resulting in low affinity of the repressor for the operator
B	Repressor has low affinity for operator	Arginine binds to the repressor and activates it, causing it to bind to the operator
C	Continuous transcription of genes to synthesize arginine	Causes bend in the promoter such that RNA polymerase has increased affinity for the promoter
D	No synthesis of genes to break down arginine	Transcription of genes to synthesize enzymes to break down arginine

18. Red-green colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, **g**, is recessive to the allele for normal colour vision, **G**.

Complete colour blindness is controlled by a different gene which is not on the X chromosome. The allele for the development of normal cones (pigment cells in the retinal layer of the eye), **B**, is dominant to the allele for no cone development, **b**.

The figure below shows the phenotypes of members of a family in which both types of colour blindness occur.



Which of the following are possible genotypes for individuals P, Q and S?

	P	Q	S
A	$BBX^G X^G$	$bbX^G Y$	$BBX^g Y$
B	$BbX^G X^g$	$BbX^g Y$	$BbX^G Y$
C	$BBX^g X^g$	$bbX^G Y$	$BbX^g Y$
D	$BbX^G X^g$	$BbX^g Y$	$BBX^G Y$

19. In *Drosophila melanogaster*, there are two known independently assorting genes **B** and **P** that affect wing shape. Flies homozygous for the recessive **b** mutant allele at one gene have bent wings. Flies homozygous for the recessive **p** mutant allele have no wings at all, regardless of their genotype at other loci. All other genotypes give normal wings. If a true breeding bent winged fly is crossed with a fly with no wings that is homozygous for the normal **B** allele and the **p** allele, what proportion of the F₂ would have no wings?

- A 1/16
- B 1/4
- C 3/16
- D 9/16

20. When a pure-breeding blue budgerigar is crossed with a pure-breeding yellow budgerigar, all the budgerigars of the F₁ generation are green. When the F₁ generation was crossed, there were 40 progeny in the F₂ generation. 3 were white, 22 were green, 8 were yellow and 7 were blue.

What are the genotypes of the parental generation?

- A AaBb x AaBb
- B aaBb x Aabb
- C aaBB x Aabb
- D aabb x AABB

21. A maize plant with green seedlings and opaque leaves was crossed with a maize plant with yellow seedlings and bright leaves. The table below shows the phenotypes of the offspring.

Green and opaque leaves	240
Green and bright leaves	36
Yellow and opaque leaves	24
Yellow and bright leaves	220

Which statement best explains these results?

- A The genes for seedling colour and leaf brightness are linked and crossing over has taken place in 11.5% of the offspring.
- B The genes for seedling colour and leaf brightness are on different chromosomes and crossing over has taken place in 11.5% of the offspring.
- C The genes for seedling colour and leaf brightness are linked and 53.8% of the offspring contain at least 1 dominant allele.
- D The genes for seedling colour and leaf brightness are on different chromosomes and 6.9% of the offspring express only two recessive alleles.

22. The table below shows the results of an early investigation into the genetic control of phenotypic variation. The dry masses of 5493 bean seeds collected from many plants were classified into nine categories.

Mass of bean/mg	51-150	151-250	251-350	351-450	451-550	551-650	651-750	751-850	851-950
Number of beans	5	38	370	1676	2255	928	187	32	2

Which statement correctly describes these data and could account for the variation shown?

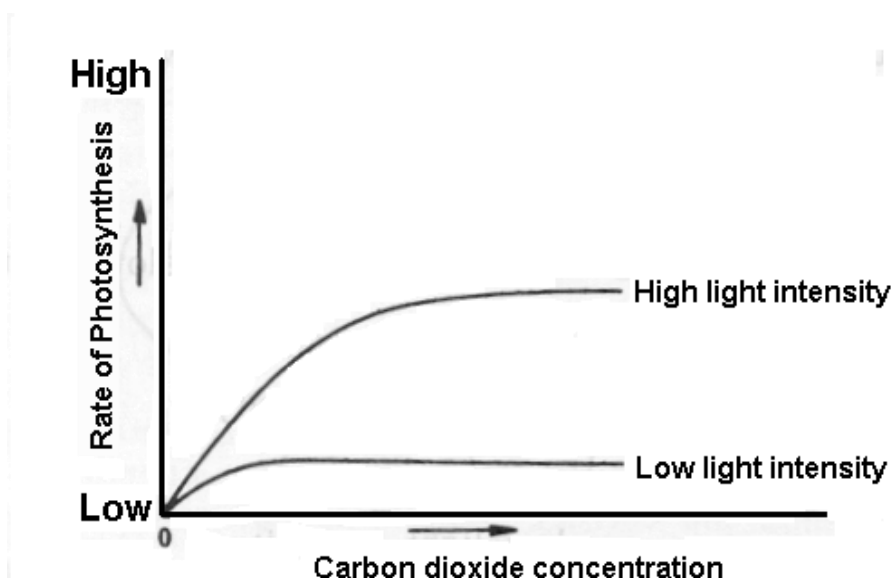
- A The phenotypic variation is continuous and could be the result of two non-linked genes acting on their own.
- B The phenotypic variation is continuous and could be the result of several non-linked genes acting on their own.
- C The phenotypic variation is discontinuous and could be the result of two linked genes acting on their own.
- D The phenotypic variation is discontinuous and could be the result of several linked genes acting on their own.

23. In the presence of a signal protein, the Ras protein, which is a G-protein, becomes activated and switches on other proteins. These in turn switch on genes that lead to cell growth and division. When the Ras gene is mutated, the Ras protein will prevent GTP hydrolysis, resulting in unintended and overactive signaling inside the cell. This happens even in the absence of incoming signals, leading to increased cell growth and division.

Which of the following describes the type of mutation in the Ras gene?

- A Gain of function, point mutation
- B Loss of function, chromosomal translocation
- C Gain of function, chromosomal translocation
- D Loss of function, point mutation

24. The graph below shows how various factors affect the rate of photosynthesis.



It can be deduced from the data shown in the graph above that the rate of photosynthesis

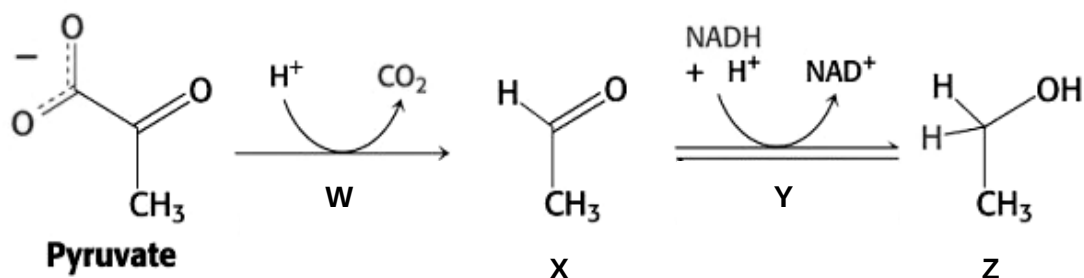
- A varies directly with carbon dioxide concentrations.
- B varies directly with light intensity.
- C is dependent on light intensity at high carbon dioxide concentrations.
- D is unaffected by carbon dioxide concentration at low light intensity.

25. The light dependent reaction consists of cyclic and non-cyclic photophosphorylation. Which statement(s) about photophosphorylation is false?

- I. It takes place on the thylakoid membrane.
- II. NAD^+ is the final electron acceptor for non-cyclic photophosphorylation.
- III. ATP is synthesized during both cyclic and non-cyclic photophosphorylation.
- IV. Photolysis takes place during both cyclic and non-cyclic photophosphorylation.

- A I only
- B II only
- C I and III
- D II and IV

26. The diagram below shows what happens to pyruvate in the root cells of plants in waterlogged soil.



Which of the following correctly explains the reaction above?

	W	X	Y	Z
A	pyruvate dehydrogenase	lactate	alcohol dehydrogenase	ethanol
B	pyruvate decarboxylase	ethanal	alcohol dehydrogenase	ethanol
C	pyruvate decarboxylase	ethanol	alcohol dehydrogenase	ethanal
D	pyruvate dehydrogenase	ethanal	alcohol dehydrogenase	ethanol

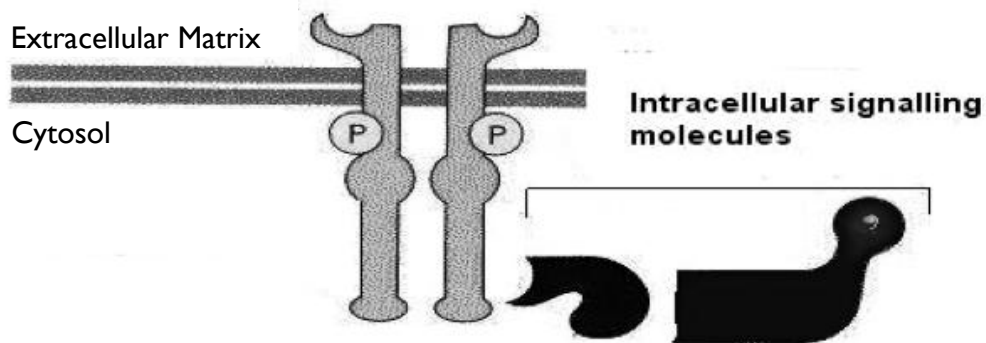
27. In aerobic respiration, ATP is synthesized by substrate level phosphorylation during glycolysis and the Krebs cycle and by oxidative phosphorylation involving hydrogen carriers.

2.5 molecules of ATP can be made from each reduced NAD and 1.5 molecules of ATP from each reduced FAD.

What is the maximum number of ATP molecules that can be made during the Krebs cycle from two molecules of glucose?

- A 18
- B 26
- C 36
- D 56

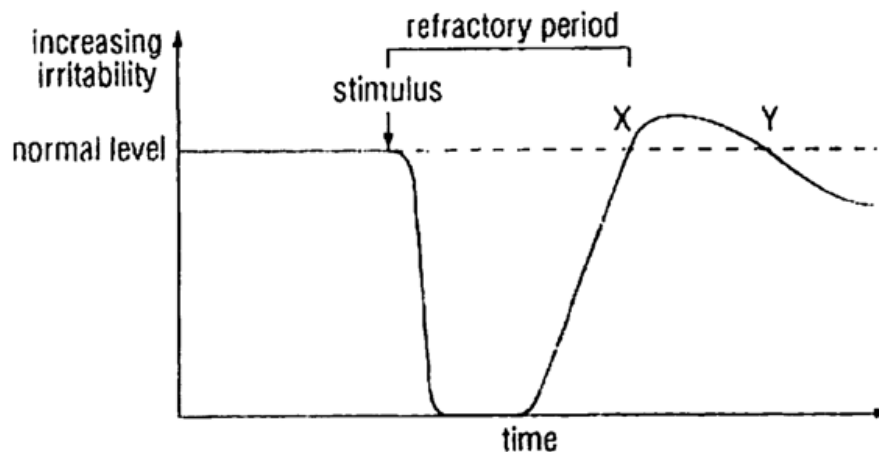
28. The diagram below shows a tyrosine kinase receptor on a liver cell.



What will be the effects when an insulin molecule binds?

	Glycogenolysis	Glycogenesis	Gluconeogenesis
A	increase	increase	increase
B	increase	decrease	increase
C	decrease	increase	decrease
D	decrease	decrease	decrease

29. The figure below shows the relative irritability of a stimulated nerve fibre.



What would happen if a second stimulus is applied during period XY?

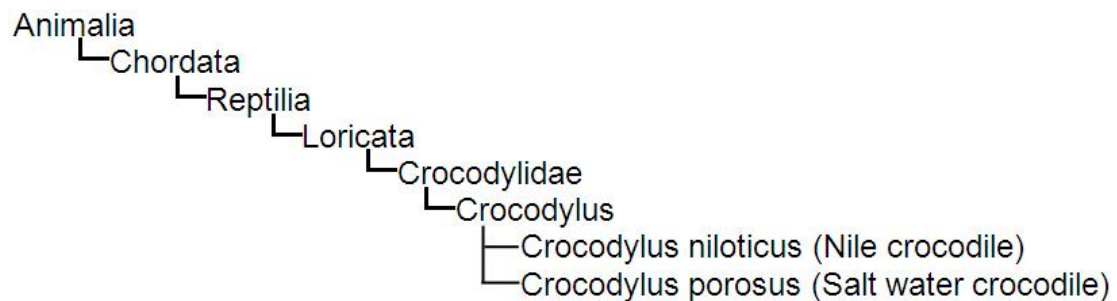
- A** The action potential will take a longer time to be generated.
- B** The reaction to the stimulus will be greater than normal.
- C** The nerve fibre will respond repeatedly to a single stimulus.
- D** A stimulus below the normal threshold will cause an action potential.

30. What is the correct sequence of events in the signal transduction pathway when insulin interacts with receptor tyrosine kinases?

- I. Activation of tyrosine kinase in tails
- II. Dimerisation of receptor polypeptides
- III. Autophosphorylation of tyrosine molecules on the cytoplasmic tail
- IV. Activation of series of relay proteins

- A** I, II, III, IV
- B** II, III, I, IV
- C** II, I, III, IV
- D** I, III, II, IV

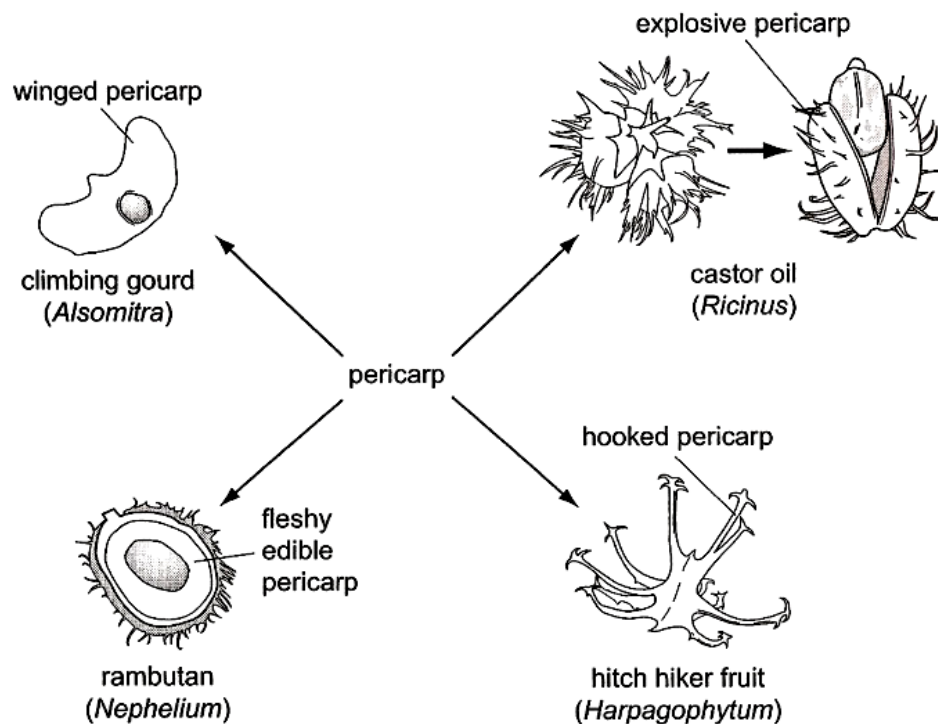
31. The chart below shows the hierarchical classification of two species of crocodiles.



To which class do these crocodiles belong?

- A** Chordata
- B** Crocodylidae
- C** Loricata
- D** Reptilia

32. The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.



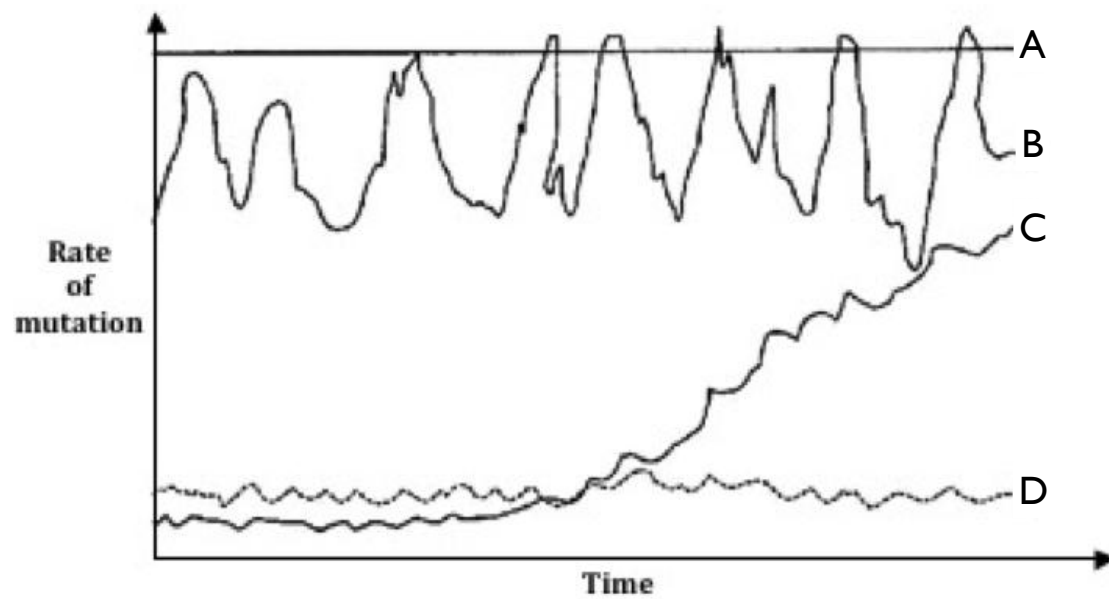
How do these examples support Darwin's theory of natural selection?

- A They illustrate how analogous structures have become adapted to perform different functions.
- B They illustrate how groups with separate evolutionary origins differ over time as a result of divergent evolution.
- C They illustrate how biotic factors operating on a variable genotype can bring about evolutionary change.
- D They illustrate the principle of adaptive radiation suggesting the divergent evolution of homologous structures.

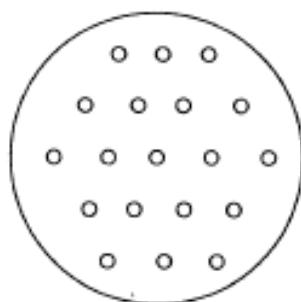
33. The average mutation rate of a gene can be used as a molecular clock. Which type of mutation would cause an underestimate of the time that has elapsed since two related species diverged from their common ancestor?

- A a mutation to the first base of a nucleotide codon
- B a mutation to the third base of a nucleotide codon
- C a mutation that returns a mutated nucleotide to its original state
- D a mutation that substitutes a different amino acid in place of the original amino acid

34. Which pattern of mutation rate would be most helpful if one desires to use a gene as a molecular clock to determine evolutionary relatedness of species that are closely related to each other?

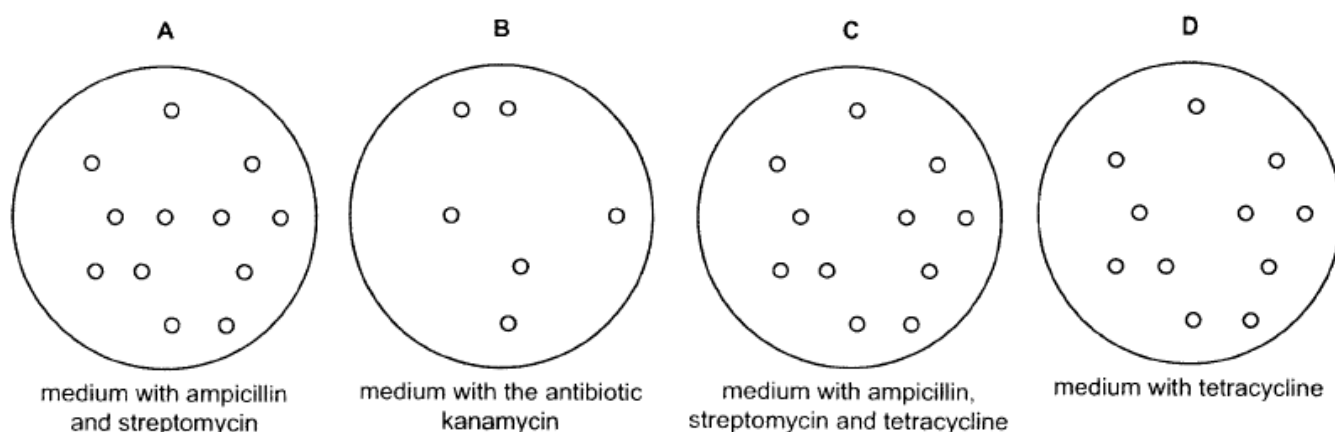


35. A gene coding for the production of a human gene product was inserted into a plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E.coli* and the bacteria grown on a nutrient medium. The resulting master plate is shown in the diagram.



Transformed cells were selected by replica plating the bacteria colonies onto media containing various antibiotics.

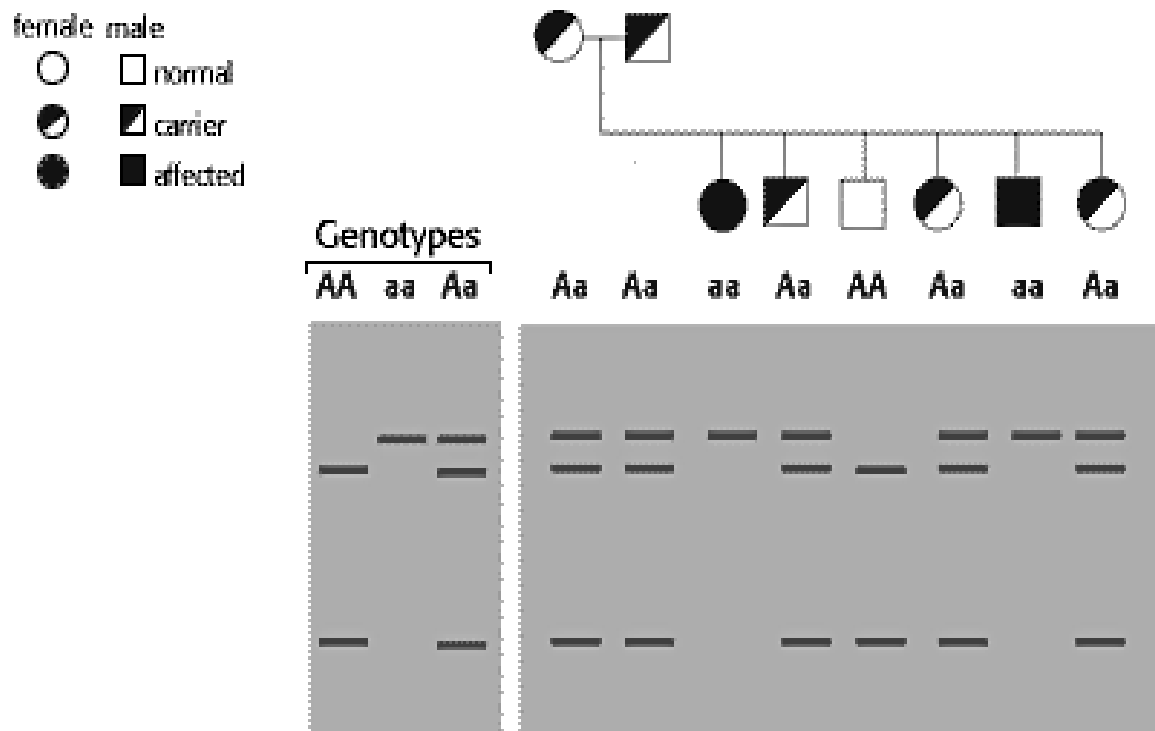
Which plate contains the colony of bacteria into which the human gene has been successfully inserted?



36. The construction of a cDNA library involves a number of stages. What is the correct sequence of stages?

	mRNA is isolated from cells	inserted into a vector and cloned	mRNA is converted into single-stranded cDNA	single-stranded cDNA is hybridized with a primer
A	1	2	4	3
B	2	1	3	4
C	3	4	2	1
D	1	4	2	3

37. Sickle-cell anemia is an autosomal, recessive human disease. A hypothetical pedigree for parents, each heterozygous for the sickle-cell allele, is shown in the figure below. The inheritance of the normal and sickle-cell allele in this family can be followed by RFLP analysis. Each of the different genotypes, homozygous normal (**AA**), heterozygous carrier (**Aa**), and homozygous recessive with sickle-cell anemia (**aa**) can be identified by RFLP analysis.



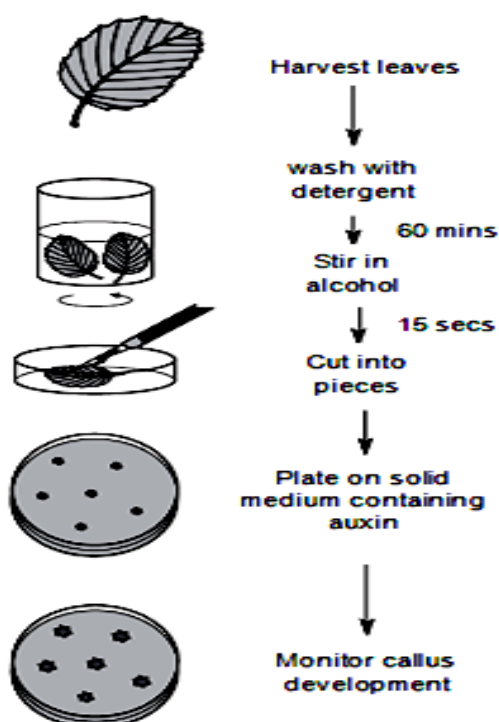
Based on the above analysis, which of the following statements is true?

- A The sickle-cell anaemia is caused by a mutation resulting in a gain of restriction site.
- B The sickle-cell anaemia is caused by a mutation resulting in a loss of restriction site.
- C The normal allele is cut using a different enzyme from that of the sickle-cell allele, hence resulting in different number of fragments produced.
- D The normal allele is a result of cutting at more than three restriction sites.

38. A patient suffers from Marfan Syndrome, an autosomal dominant genetic disorder in the gene coding for fibrillin-1 (FBN1). Almost all cells express FBN1 as its product is involved in the maintenance of connective tissues, which are abundant throughout the body. Which of the following treatment would permanently cure this patient?

- A Somatic cell gene therapy to introduce a dominant allele in a non-specific location
- B Somatic cell gene therapy to insert the functional allele in a non-specific location
- C Germ-line cell therapy to replace the faulty FBN1 gene
- D None of the above

39. Callus tissue is produced using leaves by the technique shown in the diagram below. Using this, a student investigated the effect of two different concentrations of auxin on the volume of callus material produced. The table shows the results obtained over five days.



Time (day)	volume of callus material (cm ³)	
	low concentration of auxin added	high concentration of auxin added
0	1.0	1.0
1	1.3	1.7
2	1.9	4.1
3	3.0	8.3
4	3.9	10.6
5	5.4	12.9

Which of the following statements about the experiment is false?

- I. The leaves are disinfected and sterilized by washing with detergent and alcohol.
- II. The sterilization process can alternatively be done using an autoclave machine.
- III. The only nutrient required to stimulate callus formation is auxin.
- IV. Continued subculturing at three- to four-week intervals of small cell clusters taken from calli can maintain the callus cultures for long periods of time.

- A II only
 B I and IV only
 C III only
 D II and III only

40. Salmon has been genetically modified to increase its yield within a shorter period of time. Which of the following is incorrect?

- A** Growth hormone gene from another species of salmon is inserted to greatly enhance growth.
- B** Gene coding for antifreeze protein is inserted to allow salmon to grow faster despite the cold waters.
- C** Transgenic salmon are allowed to mate with wild fish to produce stable transgenic lines in fish hatcheries.
- D** The increase in yield of transgenic salmon is due to its ability to grow throughout the year and increased efficiency in feed conversions.

1	A	21	A
2	A	22	B
3	B	23	A
4	B	24	C
5	C	25	D
6	B	26	B
7	B	27	C
8	D	28	C
9	C	29	D
10	A	30	C
11	C	31	D
12	C	32	D
13	D	33	C
14	D	34	A
15	B	35	A
16	B	36	D
17	B	37	B
18	C	38	D
19	B	39	D
20	C	40	B